ABSTRACT

VoteSecure is a WebApp voting management system developed to modernize and secure the electoral process through the integration of face recognition. The system is designed to address the vulnerabilities of traditional voting methods, such as identity fraud, unauthorized access, and lack of real-time transparency. Built using modern web technologies like Node.js, and MongoDB, VoteSecure offers a secure and efficient platform for voter authentication, vote casting, and result computation. At its core, the system employs face recognition to authenticate users before allowing them to vote, thereby eliminating multiple voting attempts and enhancing overall election integrity. Additional features include real-time result tracking, role-based access control for administrators, and secure encrypted vote storage. The application prioritizes scalability, usability, and data privacy, making it suitable for institutional, organizational, and governmental elections. While the development process involved challenges such as integrating biometric APIs, managing secure data flow, and optimizing authentication accuracy, these were addressed using agile methodologies and modular design practices. This abstract highlight the system's objective, architecture, and contributions toward creating a transparent, tamper-proof digital voting solution, positioning VoteSecure as a forward-thinking innovation in the field of secure e-governance.